Tables of Hilbert functions of filtered graphical zonotopal algebras

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Supplement to the paper *Hilbert functions of filtered graphical zonotopal algebras*

B Selected numerical tables

These tables give the Hilbert functions $\mathcal{H}_G^f$ of the filtered external zonotopal algebras $\mathcal{C}_G^f$ for a number of graphs G and nondegenerate polynomials f. Statement and equation numbers refer to the first arXiv version of the paper; Tables B1–B4 were Tables 12–15 of its version of 21 October 2026. All entries have been verified by exact rational arithmetic directly in the square-free edge algebra Φ_G, using the generators $Y_v = f(X_v)$: the script `check_tables.py` in this directory reads the edge lists from the source of the drawings below and recomputes every entry. The file `tables.json` (and `tables.csv`) lists every entry with its edge list.

If $f = u$, i.e. when $\mathcal{C}_G^f$ is the usual graded zonotopal algebra, the answer is given by Theorem 2.10. In particular, if G is a tree with n edges, then $\mathcal{H}_G^u = (1+t)^n$. We assume that $\deg f \le \Delta(G)$, since, as explained in Remark 3.3, the terms of f of degree higher than $\Delta(G)$ do not affect $\mathcal{C}_G^f$.

For a nonzero quadratic coefficient and a graph whose loopless part has no isolated vertices or isolated edges, Proposition 3.13(iii) gives $h_1 = |V|$. For graphs of maximum degree three the tables give complete profiles; for maximum degree four they give values at nine parameters.

B.1 Graphs with at most five vertices

The tables use the actual generating function $Y_v = f(X_v)$. In particular, the parameter f is not the inverse series used in the defining relations; compare (6.5). The graphs are determined by their drawings: each drawn segment or arc is one edge.

B.1.1 Graphs with four or five vertices and maximum degree three

Tables B1–B3 give the complete Hilbert profile of each displayed graph on the cubic parameter plane (Remark 6.13). Every $f = u + \alpha u^2 + \beta u^3$ has one of the five listed series: the first two columns cover $\alpha = 0$ (after scaling, $f = u$ or $f = u + u^3$), the third is the generic value for $\alpha \neq 0$, and the last two are the values on the parabolas $\beta = 2\alpha^2$ (inverse quadratics) and $3\beta = 2\alpha^2$ (exponential jets). Entries that differ from the generic column are in bold; no other value of β/α^2 is exceptional for these graphs. Each entry lists $(h_0, \dots, h_s)$ for $\mathcal{H}_G^f = \sum h_i t^i$.

B.1.2 Simple graphs with five vertices and maximum degree four

Table B4 records the Hilbert functions at nine parameters for each displayed graph. The columns f_2, f_5, f_6, f_8, f_9 lie in the generic cell of Theorem 6.10 and agree for every graph in the table, while $f_7 = u + u^2 + u^3$ lies on $[u^4]f^{-1} = 0$ (cell 7) and differs from them for seven of the eleven graphs. The columns f_1, f_3, f_4 have $\alpha = 0$. Unlike the previous tables, these columns are not shown to exhaust the Hilbert strata of these graphs.

G	u	$u+u^3$	$u+u^2$	$u+u^2+2u^3$	$u+\frac{u^2}{2}+\frac{u^3}{6}$
	$(1,3,3,1)$	$(1,4,3)$	$(1,4,3)$	$(1,4,3)$	$(1,4,3)$
	$(1,3,5,4,1)$	$(1,4,6,3)$	$(1,4,7,2)$	$(1,4,7,2)$	$(1,4,7,2)$
	$(1,3,6,8,5,1)$	$(1,4,9,8,2)$	$(1,4,10,9)$	$(\mathbf{1},\mathbf{4},\mathbf{9},\mathbf{10})$	$(1,4,10,9)$
	$(1,3,6,10,11,6,1)$	$(1,4,9,12,8,3,1)$	$(1,4,10,14,5,3,1)$	$(\mathbf{1},\mathbf{4},\mathbf{9},\mathbf{15},\mathbf{5},\mathbf{3},\mathbf{1})$	$(1,4,10,14,5,3,1)$

Table B1: Complete cubic Hilbert profiles of the connected simple graphs with four vertices and $\Delta = 3$.

G	u	$u+u^3$	$u+u^2$	$u+u^2+2u^3$	$u+\frac{u^2}{2}+\frac{u^3}{6}$
	$(1,3,4,3,1)$	$(1,4,5,2)$	$(1,4,7)$	$(1,4,7)$	$(1,4,7)$
	$(1,3,4,3,1)$	$(1,4,5,2)$	$(1,4,7)$	$(1,4,7)$	$(1,4,7)$
	$(1,3,5,5,3,1)$	$(1,4,7,5,1)$	$(1,4,9,4)$	$(1,4,9,4)$	$(1,4,9,4)$
	$(1,3,5,6,4,1)$	$(1,4,8,6,1)$	$(1,4,9,6)$	$(\mathbf{1},\mathbf{4},\mathbf{8},\mathbf{7})$	$(1,4,9,6)$
	$(1,3,6,7,4,1)$	$(1,4,8,7,2)$	$(1,4,10,7)$	$(\mathbf{1},\mathbf{4},\mathbf{9},\mathbf{8})$	$(\mathbf{1},\mathbf{4},\mathbf{9},\mathbf{8})$
	$(1,3,6,9,8,4,1)$	$(1,4,10,11,5,1)$	$(1,4,10,15,2)$	$(\mathbf{1},\mathbf{4},\mathbf{9},\mathbf{15},\mathbf{3})$	$(\mathbf{1},\mathbf{4},\mathbf{10},\mathbf{14},\mathbf{3})$

Table B2: Complete cubic Hilbert profiles of connected multigraphs with four vertices and $\Delta = 3$.

G	u	$u+u^3$	$u+u^2$	$u+u^2+2u^3$	$u+\frac{u^2}{2}+\frac{u^3}{6}$
	$(1,4,6,4,1)$	$(1,5,7,3)$	$(1,5,10)$	$(1,5,10)$	$(1,5,10)$
	$(1,4,9,10,5,1)$	$(1,5,11,10,3)$	$(1,5,14,9,1)$	$(\mathbf{1},\mathbf{5},\mathbf{13},\mathbf{10},\mathbf{1})$	$(1,5,14,9,1)$
	$(1,4,8,9,5,1)$	$(1,5,12,9,1)$	$(1,5,13,9)$	$(\mathbf{1},\mathbf{5},\mathbf{12},\mathbf{10})$	$(1,5,13,9)$
	$(1,4,8,9,5,1)$	$(1,5,10,9,3)$	$(1,5,13,8,1)$	$(\mathbf{1},\mathbf{5},\mathbf{12},\mathbf{8},\mathbf{2})$	$(\mathbf{1},\mathbf{5},\mathbf{13},\mathbf{7},\mathbf{2})$
	$(1,4,10,17,15,6,1)$	$(1,5,15,22,11)$	$(1,5,15,26,7)$	$(\mathbf{1},\mathbf{5},\mathbf{14},\mathbf{25},\mathbf{9})$	$(\mathbf{1},\mathbf{5},\mathbf{15},\mathbf{24},\mathbf{9})$
	$(1,4,9,14,13,6,1)$	$(1,5,13,19,10)$	$(1,5,14,23,5)$	$(\mathbf{1},\mathbf{5},\mathbf{13},\mathbf{21},\mathbf{8})$	$(\mathbf{1},\mathbf{5},\mathbf{14},\mathbf{22},\mathbf{6})$
	$(1,4,10,16,14,6,1)$	$(1,5,14,20,11,1)$	$(1,5,15,26,5)$	$(\mathbf{1},\mathbf{5},\mathbf{14},\mathbf{23},\mathbf{9})$	$(\mathbf{1},\mathbf{5},\mathbf{15},\mathbf{24},\mathbf{7})$
	$(1,4,10,19,25,19,7,1)$	$(1,5,15,29,26,9,1)$	$(1,5,15,33,28,3,1)$	$(\mathbf{1},\mathbf{5},\mathbf{14},\mathbf{29},\mathbf{33},\mathbf{3},\mathbf{1})$	$(1,5,15,33,28,3,1)$

Table B3: Complete cubic Hilbert profiles of the connected simple graphs with five vertices and $\Delta = 3$.

G	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	f_9
	$(1,5,6,4)$	$(1,5,10)$	$(1,5,7,3)$	$(1,5,7,3)$	$(1,5,10)$	$(1,5,10)$	$(1,5,10)$	$(1,5,10)$	$(1,5,10)$
	$(1,5,9,9,4)$	$(1,5,13,8,1)$	$(1,5,10,9,3)$	$(1,5,10,9,3)$	$(1,5,13,8,1)$	$(1,5,13,8,1)$	$(1,5,13,8,1)$	$(1,5,13,8,1)$	$(1,5,13,8,1)$
	$(1,5,12,15,12,4)$	$(1,5,15,21,7)$	$(1,5,13,16,11,3)$	$(1,5,13,16,11,3)$	$(1,5,15,21,7)$	$(1,5,15,21,7)$	$(1,5,15,21,7)$	$(1,5,15,21,7)$	$(1,5,15,21,7)$
	$(1,5,11,15,12,4)$	$(1,5,14,25,3)$	$(1,5,14,19,9)$	$(1,5,14,20,8)$	$(1,5,14,25,3)$	$(1,5,14,25,3)$	$(1,5,14,25,3)$	$(1,5,14,25,3)$	$(1,5,14,25,3)$
	$(1,5,13,21,23,15,4)$	$(1,5,15,33,28)$	$(1,5,15,28,25,8)$	$(1,5,15,29,27,5)$	$(1,5,15,33,28)$	$(1,5,15,33,28)$	$(1,5,15,32,29)$	$(1,5,15,33,28)$	$(1,5,15,33,28)$
	$(1,5,14,23,23,14,1)$	$(1,5,15,32,28)$	$(1,5,15,26,25,9)$	$(1,5,15,27,25,8)$	$(1,5,15,32,28)$	$(1,5,15,32,28)$	$(1,5,15,31,29)$	$(1,5,15,32,28)$	$(1,5,15,32,28)$
	$(1,5,11,19,21,15,4)$	$(1,5,14,29,21,5,1)$	$(1,5,14,24,22,9,1)$	$(1,5,14,25,23,7,1)$	$(1,5,14,29,21,5,1)$	$(1,5,14,29,21,5,1)$	$(1,5,14,28,22,5,1)$	$(1,5,14,29,21,5,1)$	$(1,5,14,29,21,5,1)$
	$(1,5,13,26,34,33,18,4)$	$(1,5,15,35,56,18,4)$	$(1,5,15,33,45,29,6)$	$(1,5,15,34,47,28,4)$	$(1,5,15,35,56,18,4)$	$(1,5,15,35,56,18,4)$	$(1,5,15,34,56,19,4)$	$(1,5,15,35,56,18,4)$	$(1,5,15,35,56,18,4)$
	$(1,5,14,27,35,30,15,1)$	$(1,5,15,34,53,19,1)$	$(1,5,15,31,43,28,5)$	$(1,5,15,32,47,25,3)$	$(1,5,15,34,53,19,1)$	$(1,5,15,34,53,19,1)$	$(1,5,15,33,52,21,1)$	$(1,5,15,34,53,19,1)$	$(1,5,15,34,53,19,1)$
	$(1,5,14,30,48,50,37,13)$	$(1,5,15,35,65,64,13)$	$(1,5,15,33,55,59,28,2)$	$(1,5,15,34,58,64,20,1)$	$(1,5,15,35,65,64,13)$	$(1,5,15,35,65,64,13)$	$(1,5,15,34,62,68,13)$	$(1,5,15,35,65,64,13)$	$(1,5,15,35,65,64,13)$
	$(1,5,14,30,53,73,60,41,9,4,1)$	$(1,5,15,35,67,91,48,15,9,4,1)$	$(1,5,15,33,60,76,60,27,9,4,1)$	$(1,5,15,34,63,82,56,21,9,4,1)$	$(1,5,15,35,67,91,48,15,9,4,1)$	$(1,5,15,35,67,91,48,15,9,4,1)$	$(1,5,15,34,64,90,53,15,9,4,1)$	$(1,5,15,35,67,91,48,15,9,4,1)$	$(1,5,15,35,67,91,48,15,9,4,1)$

$$f_1 = u + u^4, \quad f_2 = u + u^2, \quad f_3 = u + u^3, \quad f_4 = u + u^3 + u^4, \quad f_5 = u + u^2 + u^4,$$
$$f_6 = u + \tfrac{1}{2}u^2 + \tfrac{1}{3}u^3, \quad f_7 = u + u^2 + u^3, \quad f_8 = u + \tfrac{1}{2}u^2 + \tfrac{1}{3}u^3 + \tfrac{1}{4}u^4, \quad f_9 = u + u^2 + u^3 + u^4.$$

Table B4: Exact Hilbert functions for connected simple graphs with five vertices and $\Delta = 4$, for the nine indicated generating functions.